Section 1

2025年12月9日 星期二 10:24

Analysis & Design of Circuits Lab. Part 2 S1.
18 November, 2025.
Lab, Charles Wu.

Reference, ADC Part 1, section 1.

Aim: Learn LT1366 op-amp. Build an inverting amplifier to make audio sounds louder. Then solve problems such as cut-off in negative regions; make a non-inverting amplifier. Finally build an audio mixer, which outputs left-hand & right-hand audio that are synchronised.

Procedure.

- Before the lab.

Op-amps are integrated circuits which have function:

$$Y = A(V^+ - V^-)$$

Internally it's a complicated circuit but we can forget about it.

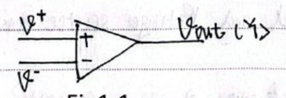


Fig1.1

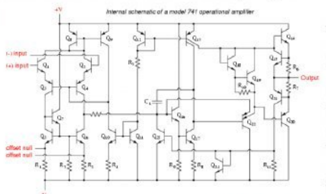


Fig1.2

When negative feedback occurs, we can assume $\left| \begin{array}{l} V^+ \simeq V^- \\ I^+ \approx I^- \approx 0 \end{array} \right.$

The op-amp we use is LT1366.

①. We need to find a gain of $-1.5 \pm 5\%$ for following inverting amp by determining Rfb:

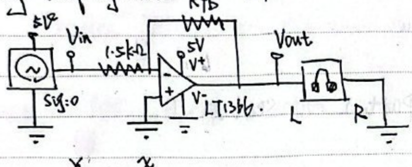


Fig1.3

$$\therefore \frac{Y}{R_{fb}} = \frac{x}{1.5k} \times (-1)$$

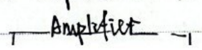
$$\frac{Y}{X} = \frac{R_{fb}}{1.5k} \times (-1) = -1.5$$

$$\therefore R_{fb} = 1.5 \times 1.5k = 2.25k\Omega \pm 5\%$$

$$= 2.1375k\Omega \sim 2.3625k\Omega$$

I choose 2.2kΩ.

②. For idealised amp, it can be represented as following:



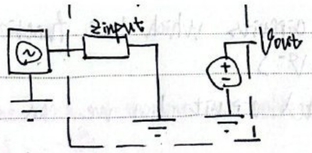


Fig1.4

which is input impedance to ground and Voltage source considered as short circuit to ground.

∵ Op-amp has infinite Z, ∴ any finite Z must generate from resistors connected to ground.

So for:

Inverting:

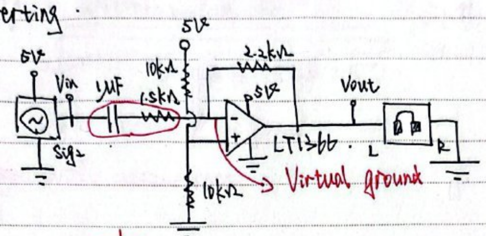


Fig1.5

$Z = R + \frac{1}{jwC}$

Eg. For $w = 2\pi f = 2\times\pi\times 2000Hz$, at 2000Hz

$Z = \sqrt{(1.5k)^2 + \frac{1}{(2\pi\times 2000\times 10^{-6})^2}} \approx 1502\Omega$.

For non-inverting

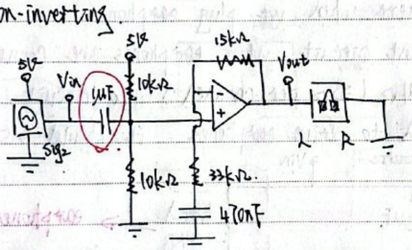


Fig1.6

$Z = \frac{1}{jwC}$

Eg. For 2000Hz, $Z = \frac{1}{2\pi\times 2000\times 10^{-6}} = 79.6\Omega$.

- Signal Source (no amp)

First, we build a signal source directly to earphones. (R on earphone adaptor to ground)

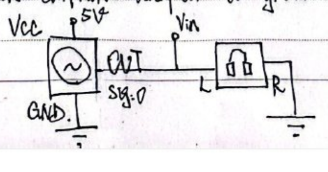


Fig1.7

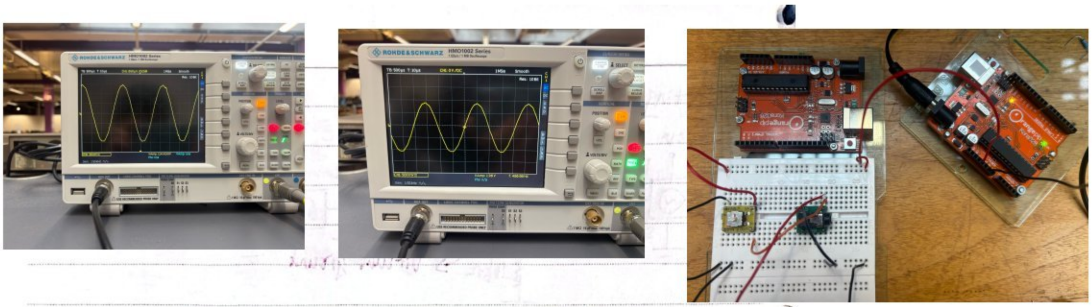


Fig1.8, 1.9, 1.10

As seen on CRO, frequency of channel ≈ 486.86Hz.

Without earphone, Vamp = 1.96V; with earphone, Vamp = 114.41 mV.

Earphone can work loudly with Vamp ≥ 500mV.

So now $114.41mV < 500mV$, without earphone it works well, but only a small sound here when we plug earphones.

Below is an equivalent circuit, when earphones are connected, we add a series 64Ω (32Ω per channel) into circuit, and V_{in} is what we need to drive earphones, it should $\geq 500mV$.

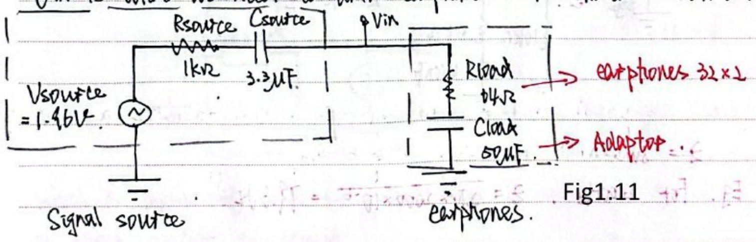


Fig1.11

$$\therefore \text{Now: } V_{in} = 1.96 \times \frac{\sqrt{64^2 + \left(\frac{1}{w \times 50 \times 10^{-6}}\right)^2}}{\sqrt{(1000+64)^2 + \left(\frac{1}{w \times C_{TOT}}\right)^2}}$$

where $w = 2\pi f = 2\pi \times 486.86$

$$C_{TOT} = \frac{3.3 \times 50}{3.3 + 50} \times 10^{-6}.$$

$$\therefore V_{in} = \frac{\sqrt{64^2 + \left(\frac{1}{2\pi \times 486.86 \times 50 \times 10^{-6}}\right)^2}}{\sqrt{(1000+64)^2 + \left(\frac{1}{2\pi \times 486.86 \times \left(\frac{3.3 \times 50}{3.3+50}\right) \times 10^{-6}}\right)^2}} \times 1.96$$

$= 0.1179V \approx 114.41mV$.

Allign with measurement.

- Inverting Amplifier.

Amp can make signal louder as:

1. It amplifies gain, $\therefore V_{amp}$ is bigger.
2. Has higher load impedance, $\therefore V_{in}$ takes greater proportion.

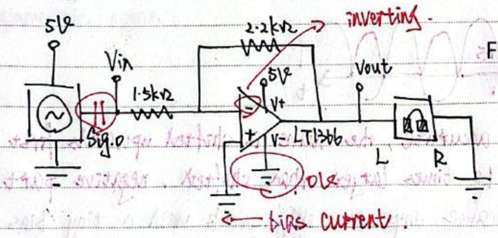


Fig1.12

The amp should increase gain by -1.5; but there're a few problems generated by this connection.

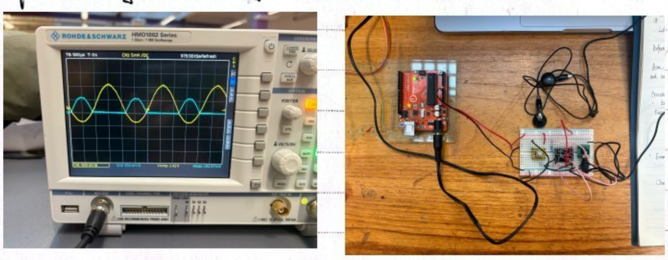


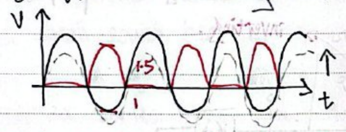
Fig1.12, 1.13

Seen on the graph, here're problems and explanations:

1. Q. Wave reflects about x-axis.
   A. It's inverting amp, so gain $= -1.5$, there's a switch on +/- $X = A(V^+ - V^-)$ and signal connects to V^-
2. Q. There're no negative voltages.

A. LT1366 is grounded and connects to 5V, ∴ output should be $0 \leq V_{out} \leq 5V$, ∴ All negative parts aren't amplified and just = 0, which is flat.

3. Q. We are not seeing a voltage 1.5 times larger.



Or to be more accurate, the wave is shifted upwards first, and output is 1.5 times larger than shifted, negative part.

A. That's because amp can only work with a tiny bias current, the bias current flows back to external circuit, so it makes a DC bias and it's no more symmetric.

To prove this, I do 3 tests:

① Connect signal directly.

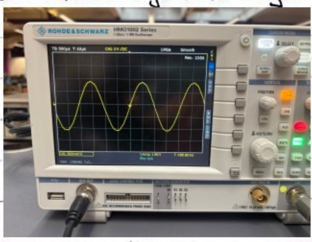


Fig1.14

which is symmetric.

② Connect to inverting amplifier, the signal (CH1) is shifted upwards.

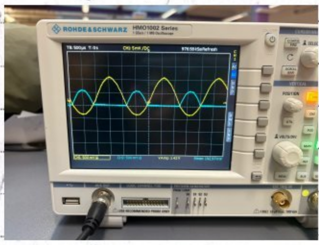


Fig1.15 Yellow Channel shows that

③ Add a capacitor before amp, as $Z_C = \frac{1}{j\omega C}$, so it will be ∞ for DC, which can block DC current from flowing, CH1 is symmetric again.

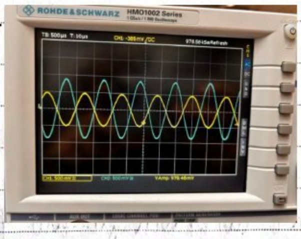


Fig1.16 Yellow Channel

· Single supply Inverting amplifier.

In order to solve problems, we make 2 changes.

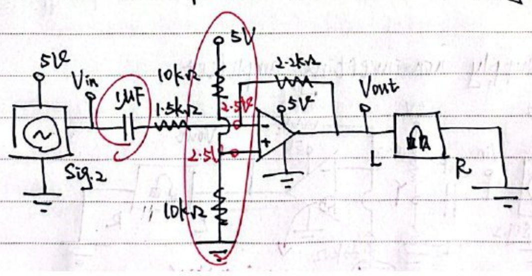


Fig1.17

①. a potential divider with 5V, so it brings a 2.5V DC bias to amp, making whole wave shown.
(we can also change ⏚ ground to -2.5V to solve problem, but that may need a 2^{nd} source).

②. A 1μF capacitor.
which is as mentioned, blocks DC bias from back to signal source.
The result is:

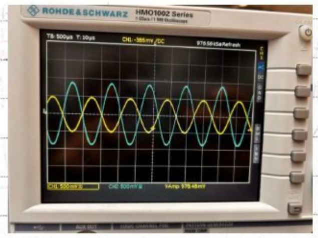
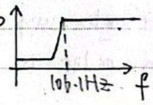


Fig1.18

Notice:
The adding capacitor makes the external circuit a high-pass filter, so it doesn't work well at low frequencies
(Under $f_c = \frac{1}{RC} \times \frac{1}{2\pi} = \frac{1}{1500 \times 1 \times 10^{-6}} \times \frac{1}{2\pi}$
$= 106.1\,Hz$).



• Single supply non-inverting amplifier.

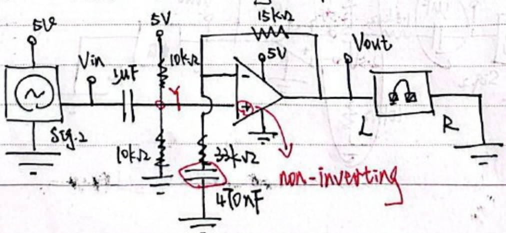
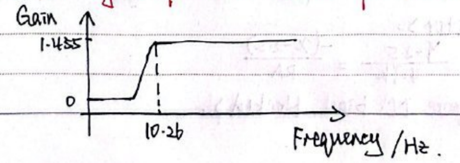
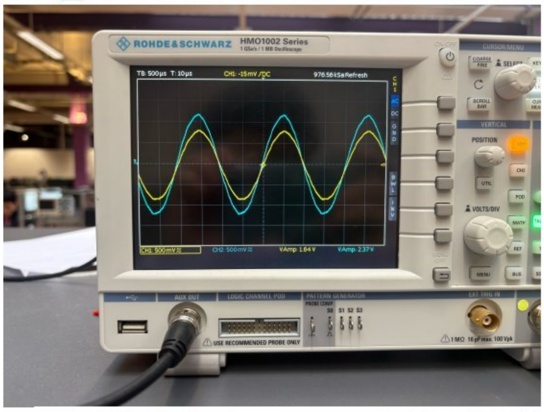


Fig1.19

Now we even try to make result better → we don't want a reflection, which is inverting any more. We build this non-inverting amp instead, and it has following changes:
①. Input and DC offset 2.5V at Y connect to V^+, as
$Y = A(V^+ - V^-)$ ⇒ Gain is positive, so non-inverting.
②. a 470nF capacitor, which prevents amplification of DC.
If no capacitor, amp will amplify everything in input, including DC bias, so the capacitor blocks DC and prevents.
(There's another way, to add a smaller bias V_{bias}, and after
Note. amplifying, $V_{out} = Gain \times V_{bias} = 2.5V$, but that's more complex.)
We know 1μF capacitor's ~~function~~ character, so we just calculate a transfer function > fc = 106.1Hz and ignore that.
$$x = Y \times \frac{Z_1}{Z_1 + Z_2}.$$
where $Z_1 = \frac{1}{j\omega C} + 33k = \frac{1}{470n \times j\omega} + 33k$.
$Z_2 = 15k$.
$\therefore$ Gain $= \frac{Y}{x} = 1 + \frac{Z_2}{Z_1} = \frac{15k}{\frac{1}{j\omega C} + 33k} = \frac{15k\,j\omega C}{1 + 33k\,j\omega C}$.
As we can see, as $\omega \to 0$, $LA = \frac{15k}{\infty + 33k} = \frac{1}{\infty} = 0$.
So will have zero gain to DC bias inputs.
as $\omega \to \infty$, $HA = 1 + \frac{15k}{0 + 33k} = 1 + \frac{15}{33} = 1.455$.
$f_c = \frac{1}{2\pi} \times W = \frac{1}{2\pi} \times \frac{1}{33k \times 470 \times 10^{-9}} = 10.26Hz$.

So will have gain of +1.455 above $f_c = 10.26Hz$; ⇒ High-pass.





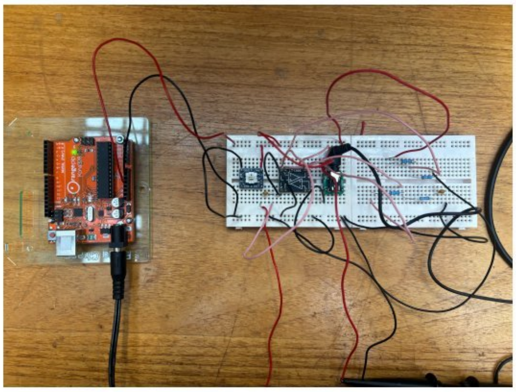


Fig1.20, 1.21

This is the best one we've done until now and it gives:

- Challenge: Audio Mixer.

We add signals 2 & 3, which are left and right parts of Turkish March together to an inverting amplifier, changing each signal's gain to make music enjoyable and not loud ($V_{out} \approx 1V$) and not has much noise.

(Note: use "SYN" to synchronise, plan to make gain of 2 < gain of 3, as main theme should be louder).

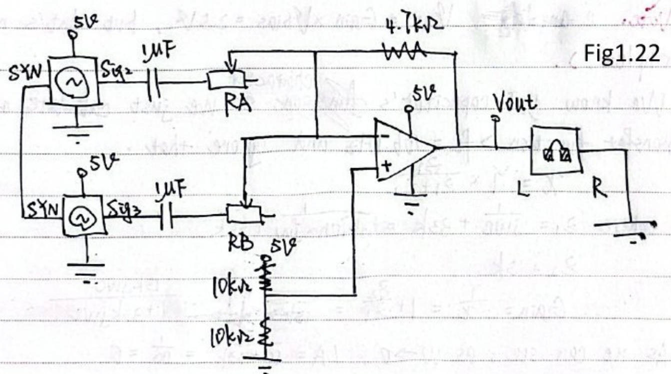


Fig1.22

We can use superposition, for a single signal:

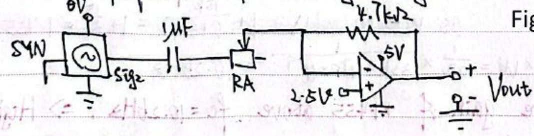


Fig1.23

(Ignore 1µF capacitor):

$$\frac{Y-2.5}{4.7k} = \frac{-(x-2.5)}{RA}$$

∴ (Ignore DC bias, blocked).

$$\frac{Y}{x} = Gain = -\frac{4.7k}{RA}$$

∵ Same for 2 signals.

$$\therefore \text{Total } V_{out} = gain_2 \cdot V_2 + gain_3 \cdot V_3 + 2.5$$

$$= -V_2 \times \frac{4.7k}{RA} - V_3 \times \frac{4.7k}{RB} + 2.5$$

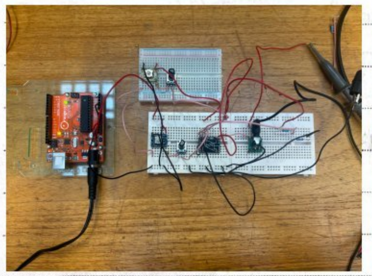
$$= 2.5 + 4.7k \times \left(\frac{-V_2}{RA} + \frac{-V_3}{RB} \right) \; V$$

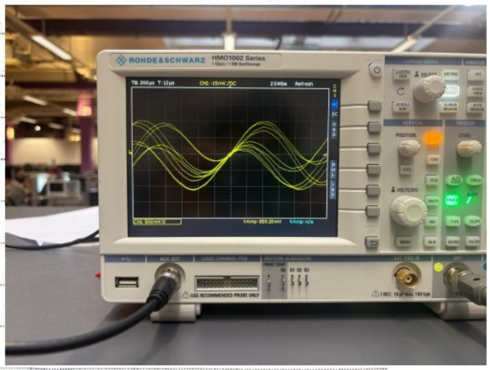
Where 2.5V is bias, the amplitude is expect to be:

$$4.7k \times \left(\frac{-V_2}{RA} + \frac{-V_3}{RB} \right) < 1V.$$

V_2 & V_3 are signals' amplitude.

And the music listens well:





$$V_{out} = -Rf\left(\frac{VA}{RA} + \frac{VB}{RB}\right).$$

Fig1.24, 1.25